

Surya Teja Tadaka

Technical Lead | Enterprise AI & Cloud Engineering | Python • AWS • Data • GenAI

suryateja233@gmail.com | +1 (914) 309-1946 | Dallas, Texas | surya-teja-tadaka.vercel.app/ | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Tech Lead with 7+ years of experience delivering AI-driven, customer-facing and cloud-native enterprise solutions. Leads end-to-end technical engagements, interfaces directly with business stakeholders, and deploys production AI/ML, automation and backend systems at scale. Strong background in Python, React, APIs, microservices and cloud platforms (AWS/GCP), with deep data engineering experience across Kafka, Glue and Redshift. Experienced in designing scalable architectures, debugging live production issues and mentoring engineering teams.

SKILLS

Programming & Backend: Python, Node.js, React, TypeScript, JavaScript, SQL, Java, FastAPI, Django, Flask, REST APIs, Webhooks, Microservices, Message Queues, SQL/NoSQL

AI / Generative AI: Generative AI, LLMs, AI Agents, Multi-Agent Orchestration, RAG, MCP, Tool Calling, Embeddings, Vector Search, Amazon Bedrock, Amazon Q Business, NLP, MLOps

Cloud & DevOps: AWS (Lambda, ECS, Fargate, S3, API Gateway, DynamoDB, IAM, Redshift, Glue, OpenSearch, Step Functions), GCP, Terraform, Infrastructure as Code, Kubernetes, Docker, CI/CD (GitHub Actions, GitLab, Jenkins), Monitoring & Alerting, Cloud Security

Data: Confluent Kafka, Streaming Pipelines, ETL, Data Pipelines, Redshift, BigQuery, Parquet, Redis, Data Quality Monitoring, Observability (Grafana, CloudWatch, DataDog)

Customer & Delivery: Technical Discovery, Scoping, Demo-driven Selling, Stakeholder Alignment, Forward Deployment, Requirements Gathering, Customer Solutioning, Project Leadership, Cross-functional Collaboration, Mentoring

PROFESSIONAL EXPERIENCE

Southwest Airlines (Qentelli) — Tech Lead

TX | June 2023 – Present

- Designed and led development of an end-to-end AI automation platform using AWS Bedrock, Amazon Q Business and multi-agent orchestration to auto-generate code, create branches and raise GitLab merge requests from Jira tickets.
- Built a multi-agent ecosystem (Jira, GitLab, Xray) through custom MCP servers enabling structured tool-calling and cross-system automation.
- Created a scalable RAG pipeline using embeddings and vector DBs to ingest and retrieve thousands of Confluence documents.
- Architected and deployed production-ready backend services and automation workflows using Python, React, FastAPI, Docker, Terraform and AWS cloud services.
- Led customer-facing technical engagements including discovery, solution design, integration planning and production rollout of AI and automation solutions.
- Acted as primary technical POC for cross-functional teams, managing stakeholders, resolving blockers and driving delivery timelines.
- Owned full SDLC including prototyping, development, CI/CD pipelines, cloud deployment, monitoring and production support; mentored junior engineers.

Smile Direct Club (Qentelli) — Sr. Software Engineer

Remote, USA | Oct 2020 – May 2023

- Led successful migration from on-prem Teradata to AWS Redshift ensuring zero data loss and optimized performance.
- Architected and deployed Confluent Kafka on AWS for real-time streaming and event-driven data processing.
- Automated AWS Glue ETL workflows to transform and store data in S3 (Parquet), improving reliability and reducing manual operations.
- Integrated AWS services including Lambda, DynamoDB, OpenSearch, Step Functions, SQS, SNS and S3 using boto3; designed batch and real-time pipelines in Python and Java.
- Built data quality monitoring using SonarQube and DataDog, reducing pipeline failures and downtime by 30%; achieved 95% automated test coverage for ETL processes.
- Converted manual infrastructure deployments to Terraform and YAML IaC, reducing environment setup time by 50%.
- Reduced cloud infrastructure cost by \$100K annually through workflow automation and test/reporting optimizations.

SS&C Technologies — Software Engineer

India | May 2019 – April 2020

- Developed backend services using Django, Flask, React and Spring Boot for claims adjudication and health insurance workflows.
- Implemented secure authentication and data protection using OAuth2, SSL, JWT and encryption.
- Applied BeautifulSoup and NLTK for parsing and text extraction, improving data accuracy and automated claims adjudication.
- Built containerized applications using Docker and CI/CD workflows using Jenkins to automate builds, tests and deployments.
- Built automated data backup pipelines to AWS S3, ensuring compliant storage of claims data.
- Reduced production defects by enforcing automated test coverage and environment parity.

SELECTED PROJECTS

Enterprise AI Code-Automation Platform

AWS Bedrock and Amazon Q Business platform with multi-agent orchestration over custom MCP servers, automating Jira → code → GitLab merge request. (Southwest Airlines)

Teradata → Redshift Migration & Streaming Platform

Zero-loss warehouse migration, Confluent Kafka streaming and automated Glue ETL into S3/Parquet; \$100K annual cloud cost reduction. (Smile Direct Club)

Cloud-Native Infrastructure as Code — github.com/TadakaSuryaTeja/aws_eks

Declarative Amazon EKS provisioning with Terraform and eksctl.

Broadcom Driver Installer — github.com/TadakaSuryaTeja/BroadcomInstaller2021

Open-source Linux hardware-enablement tool (42 stars, 27 forks).

EDUCATION

Vardhaman College of Engineering — Bachelor of Technology in Computer Science | Graduation: 2019

Government Polytechnic — Diploma in Computer Science | Graduation: 2016

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate — Amazon Web Services
- AWS Certified Cloud Practitioner — Amazon Web Services
- Machine Learning and Artificial Intelligence — AppliedAI
- Information Security-IV — NPTEL
- Mobile Application Development — NPTEL